

STRINGS - PASSWORD STRENGTH CHECKER

Topic: Strings (indexing, slicing, looping, searching, replacing, stripping)

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Course & Section: BSChE-1A

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Date: May 20, 2026

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In [1]: # =====
# PART 1: STRING BASICS - Indexing and Slicing
# =====

print("=" * 55)
print("    PASSWORD STRENGTH CHECKER")
print("=" * 55)
print("Student: Keanu Renz A. Duka (BSChE-1A)")
print("=" * 55)
print()

password = "MySecure123"
print(f"Sample password: {password}")
print()

# Indexing
print("INDEXING DEMO (using [])")
print("-" * 40)
print(f"First character: {password[0]}")
print(f"Second character: {password[1]}")
print(f>Last character: {password[-1]}")
print()

# Slicing
print("SLICING DEMO (using [start:end])")
print("-" * 40)
print(f"First 4 characters: {password[:4]}")
print(f>Last 3 characters: {password[-3:]}")
print(f"Characters 2-7: {password[2:7]}")
print()
```

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=====
      PASSWORD STRENGTH CHECKER
=====
Student: Keanu Renz A. Duka (BSChE-1A)
=====

Sample password: MySecure123

INDEXING DEMO (using [])
-----
First character: M
Second character: y
Last character: 3

SLICING DEMO (using [start:end])
-----
First 4 characters: MySe
Last 3 characters: 123
Characters 2-7: Secur

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In [2]: # =====
# PART 2: STRING LENGTH AND LOOPING
# =====

print("=" * 55)
print("STRING LENGTH AND LOOPING")
print("=" * 55)

test_word = "PASSWORD"
print(f"String: {test_word}")
print(f"Length: {len(test_word)}")
print()

# For Loop through string
print("FOR LOOP (each character):")
for letter in test_word:
    print(f"  Character: {letter}")

print()

# While Loop through string
print("WHILE LOOP (with index):")
index = 0
while index < len(test_word):
    print(f"  Index {index}: {test_word[index]}")
    index = index + 1
print()

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=====
STRING LENGTH AND LOOPING
=====
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```
String: PASSWORD
Length: 8
```

```
FOR LOOP (each character):
```

```
Character: P
Character: A
Character: S
Character: S
Character: W
Character: O
Character: R
Character: D
```

```
WHILE LOOP (with index):
```

```
Index 0: P
Index 1: A
Index 2: S
Index 3: S
Index 4: W
Index 5: O
Index 6: R
Index 7: D
```

```
In [3]: # =====
# PART 3: LOOPING AND COUNTING
# =====

print("=" * 55)
print("COUNTING CHARACTERS")
print("=" * 55)

password = "SecurePass123"
print(f"Password: {password}")
print()

# Count uppercase Letters
uppercase_count = 0
for letter in password:
    if letter.isupper():
        uppercase_count = uppercase_count + 1
print(f"Uppercase letters: {uppercase_count}")

# Count lowercase Letters
lowercase_count = 0
for letter in password:
    if letter.islower():
        lowercase_count = lowercase_count + 1
print(f"Lowercase letters: {lowercase_count}")

# Count digits
digit_count = 0
for letter in password:
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    if letter.isdigit():
        digit_count = digit_count + 1
print(f"Digits: {digit_count}")
print()

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=====
COUNTING CHARACTERS
=====

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Password: SecurePass123

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Uppercase letters: 2

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Lowercase letters: 8

```

```

Digits: 3

```

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In [4]: # =====
# PART 4: STRING CONCATENATION
# =====

print("=" * 55)
print("STRING CONCATENATION")
print("=" * 55)

a = "Super"
b = "Secure"
c = "123"
full_password = a + b + c
print(f"{a} + {b} + {c} = {full_password}")

# Adding special characters
base = "MyPassword"
suffix = "!@#"
new_password = base + suffix
print(f"{base} + {suffix} = {new_password}")
print()

```

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=====
STRING CONCATENATION
=====

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Super + Secure + 123 = SuperSecure123

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```

MyPassword + !@# = MyPassword!@#

```

```

In [5]: # =====
# PART 5: USING 'in' AS LOGICAL OPERATOR
# =====

print("=" * 55)
print("USING 'in' OPERATOR")
print("=" * 55)

password = "MySecure123"
print(f"Password: {password}")
print()

print(f"'Secure' in password? {'Secure' in password}")
print(f"'123' in password? {'123' in password}")

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print(f"'weak' in password? {'weak' in password}")
print(f"'!' in password? {'!' in password}")

if "123" in password:
    print("\n⚠ Warning: Password contains sequential numbers!")
print()
```

```
=====
USING 'in' OPERATOR
=====

Password: MySecure123

'Secure' in password? True
'123' in password? True
'weak' in password? False
'!' in password? False

⚠ Warning: Password contains sequential numbers!
```

```
In [6]: # =====
# PART 6: STRING COMPARISON
# =====

print("=" * 55)
print("STRING COMPARISON")
print("=" * 55)

password1 = "apple123"
password2 = "banana456"
print(f"Password 1: {password1}")
print(f"Password 2: {password2}")
print()

if password1 == "apple123":
    print("Password 1 is 'apple123'")

if password1 < password2:
    print(f"'{password1}' comes before '{password2}'")
else:
    print(f"'{password1}' comes after '{password2}'")
print()
```

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=====
STRING COMPARISON
=====

Password 1: apple123
Password 2: banana456

Password 1 is 'apple123'
'apple123' comes before 'banana456'
```

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In [7]: # =====
# PART 7: STRING METHODS - Lower(), upper(), find()
# =====
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print("=" * 55)
print("STRING METHODS: lower(), upper(), find()")
print("=" * 55)

password = "MySecurePassword"
print(f"Original: {password}")
print(f"lower(): {password.lower()}")
print(f"upper(): {password.upper()}")
print()

# Searching with find()
search_term = "Secure"
pos = password.find(search_term)
print(f"find('{search_term}'): {pos} (starts at position {pos})")

search_term2 = "X"
pos2 = password.find(search_term2)
print(f"find('{search_term2}'): {pos2} (returns -1 if not found)")
print()

```

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=====
STRING METHODS: lower(), upper(), find()
=====
Original: MySecurePassword
lower(): mysecurepassword
upper(): MYSECUREPASSWORD

find('Secure'): 2 (starts at position 2)
find('X'): -1 (returns -1 if not found)

```

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In [8]: # =====
# PART 8: STRING METHOD - replace()
# =====

print("=" * 55)
print("STRING METHOD: replace()")
print("=" * 55)

password = "MyOldPassword123"
print(f"Original: {password}")
print(f"replace('Old', 'New'): {password.replace('Old', 'New')}")
print(f"replace('a', '@'): {password.replace('a', '@')}")
print(f"replace('o', '0'): {password.replace('o', '0')}")
print()

```

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=====
STRING METHOD: replace()
=====
Original: MyOldPassword123
replace('Old', 'New'): MyNewPassword123
replace('a', '@'): MyOldP@ssword123
replace('o', '0'): MyOldPassw0rd123

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In [9]: # =====
# PART 9: STRIPPING WHITESPACE

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# =====

print("=" * 55)
print("STRING METHODS: strip(), lstrip(), rstrip()")
print("=" * 55)

password = "    MyPassword123    "
print(f"Original: '{password}'")
print(f"lstrip(): '{password.lstrip()}'")
print(f"rstrip(): '{password.rstrip()}'")
print(f"strip(): '{password.strip()}'")
print()
```

```
=====
STRING METHODS: strip(), lstrip(), rstrip()
=====
Original: '    MyPassword123    '
lstrip(): 'MyPassword123    '
rstrip(): '    MyPassword123'
strip(): 'MyPassword123'
```

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In [10]: # =====
# PART 10: startswith() and endswith() METHODS
# =====

print("=" * 55)
print("STRING METHODS: startswith(), endswith()")
print("=" * 55)

password = "MySecure123"
print(f"Password: {password}")
print()

print(f"startswith('My'): {password.startswith('My')}")
print(f"startswith('my'): {password.startswith('my')}") # case sensitive
print(f"endswith('123'): {password.endswith('123')}")
print(f"endswith('456'): {password.endswith('456')}")
print()
```

```
=====
STRING METHODS: startswith(), endswith()
=====
Password: MySecure123

startswith('My'): True
startswith('my'): False
endswith('123'): True
endswith('456'): False
```

```
In [11]: # =====
# PART 11: PARSING AND EXTRACTING
# From PowerPoint: "extract substring using find() and slicing"
# =====

print("=" * 55)
```

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print("PARSING AND EXTRACTING")
print("=" * 55)

# Email extraction example (from PowerPoint)
data = 'From keanu@jollibee.com Sat Jan 5 09:14:16 2024'
print(f>Data: {data}<div data-bbox="181 461 653 617" data-label="Text">


```

=====
PARSING AND EXTRACTING
=====
Data: From keanu@jollibee.com Sat Jan 5 09:14:16 2024

Position of '@': 10
Position of space after '@': 23
Extracted domain: jollibee.com

Extracted username: keanu

```


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In [12]: # =====
# PART 12: PASSWORD STRENGTH CHECKER (User Input)
# Combines multiple string concepts
# =====

print("=" * 55)
print("PASSWORD STRENGTH CHECKER")
print("=" * 55)

# Get password from user
password = input("Enter your password: ")

# Remove whitespace
password = password.strip()

print()
print("-" * 40)
print("ANALYZING PASSWORD...")

```

```
print("-" * 40)

# Check Length
length = len(password)
print(f"Length: {length} characters")

# Check for common patterns
if password.lower() == "password":
    print("⚠ WARNING: 'password' is a very weak password!")
elif password.lower() == "123456":
    print("⚠ WARNING: '123456' is a very weak password!")
elif len(password) < 8:
    print("⚠ WARNING: Password is too short (less than 8 characters)")

# Check for uppercase
has_upper = False
for letter in password:
    if letter.isupper():
        has_upper = True
        break

if has_upper:
    print("✅ Contains uppercase letters")
else:
    print("❌ No uppercase letters")

# Check for Lowercase
has_lower = False
for letter in password:
    if letter.islower():
        has_lower = True
        break

if has_lower:
    print("✅ Contains lowercase letters")
else:
    print("❌ No lowercase letters")

# Check for digits
has_digit = False
for letter in password:
    if letter.isdigit():
        has_digit = True
        break

if has_digit:
    print("✅ Contains numbers")
else:
    print("❌ No numbers")

# Check for special characters (not Letters or numbers)
has_special = False
for letter in password:
    if not letter.isalnum():
        has_special = True
        break
```

```

if has_special:
    print("✅ Contains special characters (!@#$% etc.)")
else:
    print("❌ No special characters")

# Final strength rating
print()
print("=" * 55)
print("STRENGTH RATING")
print("=" * 55)

strength_score = 0
if len(password) >= 8:
    strength_score = strength_score + 1
if has_upper:
    strength_score = strength_score + 1
if has_lower:
    strength_score = strength_score + 1
if has_digit:
    strength_score = strength_score + 1
if has_special:
    strength_score = strength_score + 1

if strength_score <= 2:
    print("🔴 WEAK - Consider using a stronger password")
elif strength_score <= 4:
    print("🟡 MEDIUM - Could be stronger")
else:
    print("🟢 STRONG - Good password!")

print("=" * 55)
print()

```

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=====
PASSWORD STRENGTH CHECKER
=====
-----
ANALYZING PASSWORD...
-----
Length: 10 characters
✅ Contains uppercase letters
✅ Contains lowercase letters
✅ Contains numbers
❌ No special characters

=====
STRENGTH RATING
=====
🟡 MEDIUM - Could be stronger
=====

```

```

In [13]: # =====
# PART 13: READING AND CONVERTING STRINGS
# From PowerPoint: "read as string, then convert using int()"

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```
# =====

print("=" * 55)
print("READING AND CONVERTING STRINGS")
print("=" * 55)

# Simulating user input (demo)
print("Demo: Converting string numbers to integers")

str_number = "250"
print(f"String: '{str_number}'")
print(f"After int(): {int(str_number)}")
print(f"After int() + 50: {int(str_number) + 50}")

# String concatenation vs numeric addition
a = "100"
b = "200"
print(f"\nString concatenation: '{a}' + '{b}' = {a + b}")
print(f"Numeric addition: int('{a}') + int('{b}') = {int(a) + int(b)}")
print()
```

```
=====
READING AND CONVERTING STRINGS
=====
Demo: Converting string numbers to integers
String: '250'
After int(): 250
After int() + 50: 300

String concatenation: '100' + '200' = 100200
Numeric addition: int('100') + int('200') = 300
```

```
In [16]: # =====
# SUBMISSION NOTE
# =====

print("=" * 55)
print("SUBMISSION COMPLETE")
print("=" * 55)
print("Name: Keanu Renz A. Duka")
print("Course: BSChE-1A")
print("Student ID: 2025-01-09327")
print("Professor: Engr. Vincent A. Calimlim")
print("Topic: Strings - Password Strength Checker")
print("=" * 55)
print("All concepts demonstrated:")
print(" - String indexing and slicing")
print(" - Looping through strings (for and while)")
print(" - String length (len)")
print(" - String concatenation (+)")
print(" - 'in' operator")
print(" - String comparison")
print(" - lower(), upper(), find(), replace())")
print(" - strip(), lstrip(), rstrip())")
print(" - startswith(), endswith())")
```

```
print(" - Parsing and extracting (find + slicing)")
print(" - Reading and converting strings")
print("=" * 55)
print("Thank you, Engr. Calimlim!")
print("=" * 55)
```

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SUBMISSION COMPLETE

=====

Name: Keanu Renz A. Duka
Course: BSChE-1A
Student ID: 2025-01-09327
Professor: Engr. Vincent A. Calimlim
Topic: Strings - Password Strength Checker

=====

All concepts demonstrated:

- String indexing and slicing
- Looping through strings (for and while)
- String length (len)
- String concatenation (+)
- 'in' operator
- String comparison
- lower(), upper(), find(), replace()
- strip(), lstrip(), rstrip()
- startswith(), endswith()
- Parsing and extracting (find + slicing)
- Reading and converting strings

=====

Thank you, Engr. Calimlim!

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In []: